Module 3.2: Cloud Computing with Google Cloud Platform (GCP)

1. Traditional E-commerce Scenario

Overview

In a traditional e-commerce setup, a company hosts its website, database, and application servers on-premises or in a local data center. The goal is to increase website traffic and sales.

Key Focus

- Increased Website Traffic: The primary objective is to attract more visitors to the e-commerce site to boost revenue.

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2. Problems Faced in Traditional E-commerce

Issues with On-Premises Infrastructure

Problem	Explanation
Increasing Cost	As traffic grows, the company needs to buy more servers, storage, and networking equipment. This leads to high capital expenditure (CAPEX) for hardware, plus

	ongoing costs for electricity, cooling, physical space, and IT staff.
Scalability Issue	Traditional infrastructure is vertically scaled (adding more power to a single server) or requires manual setup of new servers. This process is slow, prone to errors, and cannot handle sudden traffic spikes (e.g., flash sales or holidays).
Need for Fast & Fault Tolerant Systems	Customers expect websites to load quickly and remain available 24/7. Traditional setups often have single points of failure (e.g., one database server). If a server fails, the website goes down, leading to lost sales and customer dissatisfaction.

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3. Solution: Cloud Computing

Definition

- Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the internet with pay-as-you-go pricing.
- It is the use of remote servers on the internet to store, manage, and process data rather than a local server or your personal computer.

Key Characteristics

- On-demand: Resources are available when you need them, without human interaction with the service provider.
- Pay-as-you-go: You only pay for the resources you actually use (e.g., per hour, per GB of storage).
- Internet-based: All services are accessed over the internet.

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4. Cloud Services & Providers

Cloud Service Models (Implied)

While not explicitly detailed, the slides reference modular services including:

- Compute (virtual machines, serverless)
- Storage (object storage, databases)
- Data Analytics
- Machine Learning

Various Cloud Providers

- Several companies offer cloud computing services.
- The material then focuses on Google Cloud Platform (GCP) as a primary example.

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5. Why Google Cloud Platform (GCP)?

What is GCP?

- Google Cloud Platform (GCP) is a suite of cloud computing services and management tools offered by Google.
- It runs on the same infrastructure that Google uses internally for its end-user products, such as Google Search, Gmail, Google Photos, and YouTube.

Key Reasons to Choose GCP

A. Run on Google's Infrastructure

- Build applications on the same infrastructure that allows Google to:
    - Return billions of search results in milliseconds.
    - Serve 6 billion hours of YouTube video per month.
    - Provide storage for 425 million Gmail users.
- Benefits include:
    - Global Network: A high-speed, private fiber optic network connecting Google's data centers worldwide.
    - Redundancy: Multiple copies of data and services to prevent failure.
    - Innovative Infrastructure: Custom-designed servers, networking equipment, and cooling systems.

B. Scale to Millions of Users

- Applications hosted on GCP can automatically scale up to handle demanding workloads and scale down when traffic subsides. You pay only for what you use.
- Scale-up: Managed services such as App Engine or Cloud Datastore provide auto-scaling that enables your application to grow with your users – even during huge traffic spikes.
- Scale-down: Managed services also scale down automatically. You do not pay for computing resources you don't need.

C. Focus on Your Product

- Rapidly develop, deploy, and iterate your applications without worrying about system administration.
- Google manages your application, database, and storage servers so you don't have to.
- Supporting features: Managed services, Developer Tools and SDKs, Console and Administration.

Market Position

- Source: Synergy Resource Group (indicates GCP is a leading cloud provider alongside AWS and Microsoft Azure).

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6. GCP Regions and Zones

Global Footprint (2025 data)

- 41 Regions
- 124 Zones
- 187 Network Edge Locations
- 200+ Countries and Territories

Key Definitions

Regions

- A region is a specific geographical location where you can host your resources (e.g., `asia-east1`, `us-west1`).
- Regions are independent geographic areas that consist of zones.
- A region consists of three or more zones housed in three or more physical data centers (some regions are expanding to meet this).
- Example: The `us-west1` region denotes a region on the west coast of the United States that has three zones: `us-west1-a`, `us-west1-b`, and `us-west1-c`.

Zones

- A zone is a deployment area for Google Cloud resources within a region.
- Zones should be considered a single failure domain within a region.
- Best Practice: To deploy fault-tolerant applications with high availability and protect against unexpected failures, deploy your applications across multiple zones in a region.

Zonal vs. Regional Resources

Resource Type	Definition	Example

Zonal Resources	Operate within a single zone. A zonal outage can affect some or all resources in that zone.	A Compute Engine virtual machine (VM) instance resides within a specific zone.
Regional Resources	Can be used by any resource in that region, regardless of zone.	To attach a zonal persistent disk to an instance, both must be in the same zone. To assign a regional static IP address to an instance, the instance must be in the same region but can be from any zone.

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7. Creating a Free Account on GCP

Free Tier Benefits

- $300 in free credit for new customers to try Google Cloud products and build a proof of concept.
- You won't be charged until you activate your full paid account.
- Credits are valid for 90 days.
- Signup: Go to the GCP website and create an account.

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8. GCP Services In Depth

A. Compute Services

Compute Engine

- Definition: A secure and customizable compute service that lets you create and run virtual machines (VMs) on Google's infrastructure.
- Key points:
    - Virtual machines for any workload.
    - Easily create and run online VMs on high-performance, reliable cloud infrastructure.
- Free Tier (for new users):
    - One `e2-micro` VM instance per month.
    - Up to 30 GB storage.
    - Up to 1 GB of outbound data transfers per month (not charged against credits).

Other Compute Options

- App Engine: A Platform-as-a-Service (PaaS). You focus only on your code; Google handles patching and maintenance.
- Compute Engine: Infrastructure-as-a-Service (IaaS). You get raw virtual machines and flexibility to build anything.

B. Networking Services

Cloud VPC (Virtual Private Cloud)

- Definition: A global virtual network that spans all GCP regions.
- Features:
    - Network of 41 regions, 128 zones in 200+ countries with 99.99% uptime.
    - Use a single VPC to span multiple regions without communicating across the public internet.
    - Single VPC for an entire organization, isolated within projects.
    - Increase IP space with no downtime.
- Free tier: New customers get $300 in free credits to spend on VPC.

Cloud CDN (Content Delivery Network)

- Definition: A Content Delivery Network that uses Google's global edge network to serve content closer to users, accelerating websites and applications.
- Benefits: Fast, reliable web and video content delivery with global scale.
- Free tier: $300 in free credits for new customers.

What is a CDN?

- Simple Explanation: A network of interconnected servers that speeds up webpage loading for data-heavy applications.
- How it works (without CDN): When a user visits a website, data travels from the website's server across the internet to the user. If the user is far from the server, large files load slowly.
- How it works (with CDN): Website content is stored (cached) on CDN servers geographically closer to users. Content reaches users' computers much faster.
- Primary Purpose: To reduce latency (delay in communication created by a network's design).
- Benefits: Decreases web traffic to the main web server, reduces bandwidth consumption, and improves user experience.
- Analogy: A CDN is like an ATM on every corner. Having cash machines conveniently located makes getting money fast and efficient, avoiding long bank lines.

C. Storage and Databases

Cloud Storage (Object Storage)

- Definition: A managed service for storing unstructured data. You can store any amount of data and retrieve it as often as you like.
- Key Concepts:
    - Object: An immutable piece of data consisting of a file of any format (e.g., `puppy.png`).
    - Bucket: A container that holds objects. All buckets are associated with a project.
    - Project: Each project has its own set of Cloud Storage APIs and resources.
    - Organization: Your company (e.g., `exampleinc.org`) can have multiple projects.

- Workflow: Create a project → Create a bucket → Upload objects → Download objects. Grant permissions to control access.

Storage Classes

- A storage class is metadata attached to each object that affects its availability and pricing. When you create a bucket, you can set a default storage class. If none is specified, the default is Standard storage.

Storage Class	Description	Best For
Standard	Frequently accessed ("hot") data.	Websites, streaming videos, mobile apps.
Nearline	Low cost, highly durable. Data can be stored for 30 days.	Infrequently accessed data (e.g., backups accessed once a month).
Coldline	Very low cost, highly durable. Data can be stored for 90 days.	Infrequently accessed data (e.g., disaster recovery backups accessed once a quarter).
Archival	Lowest cost, highly durable. Data can be stored for 365 days.	Data archiving, online backup, disaster recovery (accessed less than once a year).

Cloud SQL

- Definition: A fully managed relational database service for MySQL, PostgreSQL, and SQL Server.
- Features: Rich extension collections, configuration flags, developer ecosystems.
- Free tier: New customers get $300 in free credits to spend on Cloud SQL.

Datastore

- Definition: A highly scalable NoSQL database for web and mobile applications.
- Features:
    - Automatically handles sharding and replication.
    - Highly available and durable; scales automatically to handle application load.
    - Provides ACID transactions, SQL-like queries, indexes, and more.

Sharding Explained

- Definition: Dividing a large database into smaller, faster, more manageable parts called "shards." Each shard is stored on a different node (server).
- Need for Sharding:
    - Without sharding: Searching a database with 100,000 student records requires scanning all 100,000 records each time → high transaction cost.
    - With sharding: Divide records by year. Each shard has 1000-5000 records. Searching is much faster and cheaper.
- How it works:
    - Data is partitioned based on a criterion (geographic location, user ID, time period).
    - Data is distributed across multiple servers/nodes.
    - A shard key (unique identifier) maps data to its corresponding shard.

Firestore

- Definition: The next generation of Datastore.
- Improvements: Introduces several improvements over Datastore.
- Migration: Existing Datastore users can create a new Firestore in Datastore mode. In the future, all existing Datastore databases will be automatically upgraded.

Cloud Bigtable

- Definition: An HBase-compatible, enterprise-grade NoSQL database service.
- Features: Single-digit millisecond latency, limitless scale, 99.999% availability for large analytical and operational workloads.
- Note: HBase is a column-oriented non-relational DBMS that runs on top of Hadoop Distributed File System (HDFS).
- Free tier: $300 in free credits for new customers.

Persistent Disk

- Definition: Reliable, high-performance block storage for virtual machine instances.
- Features: Enterprise scale, limitless flexibility, competitive price/performance.
- Free tier: $300 in free credits for new customers.

D. Big Data Services

BigQuery

- Definition: A fully managed, AI-ready data analytics platform and serverless, cost-effective enterprise data warehouse.
- Features: Works across clouds, scales with your data, includes BI, machine learning, and AI. Use SQL queries to analyze your data.

Dataproc

- Definition: A fully managed, highly scalable service for running Apache Hadoop, Apache Spark, Apache Flink, Presto, and 30+ other open-source tools.
- Features: Run open-source data analytics at scale with enterprise-grade security.

Dataflow

- Definition: A managed service for executing batch and streaming data processing patterns.
- Features: Unified stream and batch processing that is serverless, fast, and cost-effective.
- Free tier: $300 in free credits for new customers.

Cloud Life Sciences

- Definition (Originally): A suite of services for managing, processing, and transforming life sciences data (e.g., genomics, biomedical data) using containerized workflows.
- Important Deprecation Notice:
    - Cloud Life Sciences is deprecated and will no longer be available on Google Cloud after July 8, 2025.
    - Migration Path: Use cases are now supported by Batch. Users should migrate their workloads to Batch.

E. Other Service Categories

The slides list the following categories but provide no detailed content:

- Machine Learning
- Identity & Security
- Management and Developer Tools

Module 3.2: Google AppEngine

Main Heading: Google App Engine (GAE) - Overview

Definition

Google App Engine (often referred to as GAE or simply App Engine) is a cloud computing Platform as a Service (PaaS). It is designed for developing and hosting web applications in Google-managed data centers.

Explanation

- As a PaaS offering, GAE abstracts away the underlying infrastructure (servers, networking, storage, operating systems).
- Developers only need to focus on writing their application code; Google handles all infrastructure management, including scaling, load balancing, and maintenance.

Key Features of Google App Engine

1. Popular Programming Languages

- Explanation: GAE supports multiple popular programming languages, allowing developers to build applications using the language they are most comfortable with.
- Supported Languages: Node.js, Java, Ruby, C#, Go, Python, or PHP.

2. Fully Managed Environment

- Explanation: A fully managed environment means that developers focus solely on writing code. App Engine automatically manages all infrastructure concerns, such as server provisioning, patching, and monitoring.

3. Powerful Application Diagnostics

- Explanation: GAE provides tools to monitor application health, performance, and error debugging.
- Tools included:
    - Cloud Monitoring: Monitors the overall health and performance of your app.
    - Cloud Logging: Records and manages log data from your application.
    - Error Reporting: Helps quickly diagnose and fix bugs by aggregating and displaying errors.

4. Application Versioning

- Explanation: Versioning allows developers to host multiple different versions of the same application simultaneously.
- Use cases:
    - Create separate environments for development, testing, staging, and production.
    - Easily roll back to a previous version if a new deployment has issues.

5. Application Security

- Explanation: GAE includes built-in security features to safeguard applications.
- Key security features:
    - App Engine Firewall: Allows defining access rules (e.g., allow/deny specific IP addresses) to control who can reach the application.
    - Managed SSL/TLS Certificates: By default, GAE provides managed SSL (Secure Socket Layer) and TLS (Transport Layer Security) certificates for custom domains at no additional cost. This ensures encrypted communication between users and the application.

6. Services Ecosystem

- Explanation: GAE integrates with a growing ecosystem of Google Cloud services.
- Examples: Developers can tap into various cloud developer tools and services (e.g., Cloud Storage, Cloud SQL, Cloud Datastore) directly from their App Engine application.

Google's Infrastructure Behind GAE

- Scale: Google has hundreds of data centers worldwide, with over 460,000 servers installed. For example, 200 data centers may be used simultaneously for various cloud applications.
- Data Storage: Data items (text, images, video) are stored and replicated across multiple servers to tolerate faults or hardware failures.
- Sandboxing & Scaling: Applications run in a sandboxed environment across multiple servers. App Engine provides automatic scaling – as the number of requests increases, more resources are automatically allocated to handle the additional demand.

Architecture of Google App Engine

Explanation

App Engine is a platform for developing scalable web-accessible applications. The platform is logically divided into four major components.

Four Major Components of GAE Architecture:

1. Infrastructure: The underlying hardware and network resources in Google data centers.
2. The Runtime Environment: The execution context for applications (e.g., Python, Java, Go runtimes).
3. The Underlying Storage: Different storage levels (in-memory cache, semi-structured data storage, long-term static storage).
4. The Set of Scalable Services: Services like Datastore, MemCache, Mail, XMPP, etc., that developers can use.

Primary Function

- GAE hosts web applications and its primary function is to serve user requests efficiently.
- For each HTTP request:
    - GAE locates the servers hosting the application that processes the request.
    - It evaluates the load on those servers.

    - If necessary, it allocates additional resources (e.g., more servers) or redirects the request to an existing, less-loaded server.

Runtime Environment

Definition

The runtime environment represents the execution context of applications hosted on App Engine.

Explanation

- The App Engine infrastructure code is always active and running.
- The runtime comes into existence when a request handler starts executing.
- The runtime terminates once the handler has completed its work.

Supported Runtimes (Languages)

Currently, three primary languages are mentioned, though modern GAE supports more:

1. Java: Supports Java Server Pages (JSP) and the Java Servlet standard for web applications.
2. Python: Uses an optimized Python interpreter. Supports the Python standard library, but modules that implement potentially harmful operations are removed. Attempting to import them generates exceptions.
3. Go: Allows applications developed with the Go programming language to be hosted and executed.

Standard Environment & Containers

- The App Engine standard environment is based on container instances running on Google's infrastructure.
- Containers are preconfigured with one of several available runtimes (e.g., Java 17, Python 3.10).
- This environment makes it easy to build and deploy applications that run reliably under heavy load and with large amounts of data.

- Applications run in a secure, sandboxed environment that is independent of the hardware, operating system, or physical location of the server.

Instance Classes

Definition

The instance class determines the amount of memory and CPU available to each instance, the amount of free quota, and the cost per hour after the app exceeds the free quota.

Important Note on Memory Limit

- The memory limit includes memory used by your app plus memory needed by the runtime itself.
- Java runtimes use more memory to run your app than other runtimes.

Table of Instance Classes

Instance Class	Memory Limit	CPU Limit	Supported Scaling Types
F1 (default)	384 MB	600 MHz	automatic
F2	768 MB	1.2 GHz	automatic
F4	1536 MB	2.4 GHz	automatic
F4_1G	3072 MB	2.4 GHz	automatic
B1	384 MB	600 MHz	manual, basic
B2 (default)	768 MB	1.2 GHz	manual, basic

B4	1536 MB	2.4 GHz	manual, basic
B4_1G	3072 MB	2.4 GHz	manual, basic
B8	3072 MB	4.8 GHz	manual, basic

- F classes: Designed for automatic scaling (frontend instances).
- B classes: Designed for manual or basic scaling (backend instances).

Runtime Lifecycle

Explanation

App Engine standard environment runtimes use open-source components maintained by their communities. Each runtime is identified by its language version (e.g., Java 17, Python 3.10).

Support Phases

1. General Availability (GA) - Supported

- Google provides full support.
- Runtime components receive regular security and bug fixes.
- Google avoids breaking changes; any breaking changes are announced in advance.

2. End of Support

- What happens:
  - Google no longer applies security updates or patches.
  - Your application continues to run and receive traffic.
  - You cannot create new applications or update existing ones on this runtime.
  - Issues arising from this runtime are not eligible for technical support.

- Special case: Google may permit an organization to re-enable deployments on unsupported runtimes under specific terms.

3. Deprecated

- If re-enabling was allowed, Google removes that ability when the runtime is deprecated.
- Google makes reasonable efforts to notify users in advance via in-app notifications.

4. Decommissioned

- Applications still using a decommissioned runtime may be disabled without further notice.

Notification Period

- App Engine issues in-app notifications 90 days before the application reaches the end of support.
- Developers are advised to upgrade to a newer runtime.

Runtime Lifecycle Table Summary

Action / Feature	GA-level Support	End of Support	Deprecated	Decommissioned
Creation & Redeployment	Yes	No	No	No
Project Configuration Updates	Yes	Yes	No	No
Running existing workloads	Yes	Yes	Yes	Yes
UI & CLI Warnings	Yes	Yes	No	No

Language patches	Automatic	No updates	No updates	No updates
Patching APIs & SDKs	Automatic	No updates	No updates	No updates
Patching OS	Automatic	No updates	No updates	No updates
Customer Support	GA-level support	No support	No support	No support

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Sandboxing

Definition

Sandboxing provides an application environment with an isolated and protected context. It allows an application to execute without causing a threat to the server and without being influenced by other applications.

Explanation

- Metaphor: Like giving each child their own sandbox to play in alone if they do not play well together.
- GAE supports applications developed with interpreted or managed languages (e.g., Python, Java, Go).
- Sandboxing is achieved by using modified runtimes that disable some common features normally available.

Disallowed Operations (Throw Exceptions)

If an application tries to perform any potentially harmful operation, an exception is thrown and execution is interrupted.

Specific disallowed operations:

- Writing to the server's file system.
- Accessing the computer through the network except when using Mail, UrlFetch, and XMPP.
- Executing code outside the scope of a request.
- Processing a request for more than 30 seconds.
- (From context) Executing code outside a queued task.

Storage in App Engine

Explanation

App Engine provides different types of storage that operate differently depending on the volatility (temporary vs. permanent) of the data.

Three Levels of Storage

1. In-memory cache (MemCache): Volatile, fast access for frequently accessed objects.
2. Storage for semi-structured data (Datastore): Persistent, highly scalable NoSQL database.
3. Long-term storage for static data (Static file servers): For files like CSS, HTML, JavaScript, images, icons, sound files.

Static File Servers

- Web applications consist of dynamic data (from logic/user interaction) and static data (graphical layout components).
- Static files are not frequently modified, so they are hosted on dedicated static file servers.

Google Cloud Platform Services (Used with GAE)

1. Datastore

- Definition: A highly scalable NoSQL database for web and mobile applications.
- Features:
    - Automatically handles sharding (distributing data across multiple servers) and replication (copying data for fault tolerance).
    - Highly available and durable database that scales automatically to handle application load.
    - Provides ACID transactions (Atomicity, Consistency, Isolation, Durability).
    - Supports SQL-like queries and indexes.
    - Uses GQL (Graph Query Language) for queries.
- Underlying Infrastructure: Based on Bigtable – a redundant, distributed, semi-structured data store that organizes data in tables.

2. MemCache

- Definition: A distributed in-memory cache service optimized for fast access.
- Explanation:
    1. Provides developers with a volatile store for frequently accessed objects.
    2. The caching algorithm automatically removes rarely accessed objects.
- Usage Pattern:
    1. Application first looks up an object in MemCache.
    2. If there is a miss (object not found), it is retrieved from Datastore.
    3. The object is then put into MemCache for future lookups.
- Benefit: Significantly reduces data access time.

3. Mail Service

- Purpose: Allows applications to send and receive emails.
- Features:
    - Send emails on behalf of the application to specific user accounts.
    - Include several types of attachments.
    - Target multiple recipients.
    - Operates asynchronously (does not block the application while sending).
    - On failed delivery, the sending address is notified via an email detailing the error.

4. XMPP (Extensible Messaging and Presence Protocol)

- Definition: An open standard for instant messaging and presence information, based on XML.

- Capabilities: Instant messaging, presence, multi-party chat, voice/video calls, collaboration, lightweight middleware, content syndication, and generalized routing of XML data.
- Use with GAE:
    - Any XMPP-supported chat service (e.g., Google Talk) can send/receive messages to/from the web application.
    - The application is identified by its own address.
    - Can be used for human interactions or to connect with chat bots or implement an administrative console.

5. Account Management (Google Accounts)

- Explanation: GAE simplifies account management by allowing developers to leverage existing Google Accounts.
- How it works:
    - Web applications can store profile settings as key-value pairs attached to a Google account.
    - When a user authenticates with their Google account, settings are quickly retrieved.
- Advantage: No need to implement a custom authentication system. Users must have a Google account, but no further implementation is needed.
- Corporate use: Particularly advantageous in corporate environments using Google Apps, as applications can integrate with all other Google services and profile settings.

6. Image Manipulation Service

- Purpose: Allows applications to perform common image processing tasks.
- Operations Supported: Resizing, rotation, mirroring, enhancement (e.g., adding watermarks, applying simple filters).
- Design: Mostly designed for lightweight image processing and is optimized for speed (also used in other Google products).

Additional GAE Services & Features

Cron Jobs (Scheduled Tasks)

- Definition: The App Engine Cron Service allows configuration of scheduled tasks that operate at defined times or regular intervals.
- Common name: Cron jobs.
- Triggering: Automatically triggered by the Cron Service.
- Examples:
    - Send a report email daily.
    - Update cached data every 10 minutes.
    - Update summary information once an hour.

UrlFetch Service

- Definition: Allows a script (application) to access other resources on the web by fetching URLs.
- Capabilities:
    - Issue HTTP and HTTPS requests.
    - Receive responses.
- Underlying technology: Uses Google's network infrastructure for efficiency and scaling.
- Related Classes:
    - HTTPResponse: Allows access to specific information on HTTP responses (e.g., status code, headers, content).
    - UrlFetchApp: Fetches resources and communicates with other hosts over the Internet.

Well-Known GAE Applications

Google's own large-scale applications run on infrastructure similar to GAE. These include:

- Google Search Engine
- Google Docs
- Google Earth
- Gmail

Characteristics:

- Support large numbers of users simultaneously.

- Run in Google data centers, each containing thousands of server nodes forming different clusters. Each cluster can run multipurpose servers.
- GAE also provides Google-specific services, such as using a Gmail account directly as a login service.